

Musica Installation Guide
Version V 1.1

Installation instructions

To install Musica, follow the steps below:

Download the Musica from the GitHub project database:

(<https://github.com/jcipale/Notes/RELEASE>). Select the 'Download' icon from the github interface. The Musica_V1.1 tar-file will be downloaded into your Downloads directory¹.

WARNING for current Musica users:

If you are using a new version of Musica, you will need to rebase the your musica.db file as changes were made to the database schema or data dictionary. Before installing the Musica tarball, mv or rename the current Musica install. You will want to keep the musica.db. Once the current install has been preserved, proceed with the new installation.

Create the installation directory:

After the compressed file has been retrieved, identify a suitable location to install the application. A typical local install location is in the users home (/home/<user_id> for linux/MacOS or C:\Users\<user_id> for Windows) or on a mounted network drive. Follow the following steps to install the application:

- From the command-line, create the directory 'Musica' then change directories into the newly created 'Musica' directory.
- Copy the tar file Musica_v1.1.tar.gz to the current location.
- From a terminal window, issue the commands -
% gunzip Musica_v1.1.tar.gz
% tar xvf Musica_v1.1.tar.

1) For Linux/Mac, that would typically be /home/<user>. Windows would be C:\Users\Username.

The gz file will be uncompressed into the standalone tar-file. The tar-file will then be expanded and the target file structure created, complete with executable scripts, and database loading utilities.

```
Musica
├── admin
│   ├── musica_db_clean_slate_sql.py
│   ├── musica_db_debug.py
│   └── musica_db_sql_provision_sqlite.py
├── bin
│   ├── musica_db_add_recording_sqlite.py
│   └── musica_db_batch_load_sqlite.py
├── config
│   └── musica.conf.template
├── data
│   ├── exports
│   ├── imports
│   └── template.csv
├── db
├── docs
│   ├── create_provision.jpg
│   ├── Musica_db.jpg
│   ├── Musica.doc
│   ├── Musica_model.jpg
│   └── schema_load.jpg
├── install
│   └── musica_db_install.py
├── LICENSE
├── sql
│   ├── install
│   │   ├── Audit_Recordings_sqlite.sql
│   │   ├── Create_Schema_sqlite.sql
│   │   ├── Create_Staging_sqlite.sql
│   │   └── v_recordings_display_sqlite.sql
│   └── runtime
│       ├── Promote_Recordings_sqlite.sql
│       ├── promote_staging_sqlite.sql
│       └── recordings_load_sqlite.py
├── tests
├── tools
│   ├── Musica.py
│   ├── schema_tmp.csv
│   └── template.json
├── VERSION
└── web
```

The folder 'web' is an optional directory and is used to contain/install the Notes front-end/browser-based UI. For more information on Notes installation and usage, refer to the Notes Installation Guide or the Musica-Notes Reference Guide.

System Setup:

Since Musica is a cross-platform application, there are some subtle differences between Windows and linux/unix/MacOS:

- Python IS NOT typically installed by default on Windows and the install script will not install them as part of the application setup. Users may need to install Python 3.11.9 if the python interpreter is not already in place. Go to <https://www.python.org/downloads/windows/>. Select Python 3.11.9 for download/install.

- SQLite3 is not installed on Windows. Users will need to install the SQLite3 executable on their local host. Go to <https://sqlite.org/download.html>. The download page references the SQLite3 command-line interface tool that will be used for some features of the application. Select and download the SQLite-tools-win-X64-3510200.zip.
- sqlite3 is an optional tool on all unix-based platforms (unix, linux, MacOS) and depending on the flavor in use, will need to be retrieved by download (apt, yum, pacman). See the documentation for your platform to use the correct tool.

From the Download folder, unzip the python311.19 file and install it. This will install Python 3.11.9 in the Local\Programs folder of the user's account. After the python interpreter has been installed, copy the SQLite3 tarball to your root account (C:\Users\<username>) then install the application.

After each application has been installed, edit the %PATH% environment variable (from the keyboard, type Win + R then enter sysdm.cpl in the dialog box and press "Enter", then Select "Advanced". After the 'Advanced' tab opens, look for "Environment Variables" in the lower-right corner of the popup window. In the top of the 'Environment Variables' dialog will be the list of current Environment Variables. Select the value labeled 'Path' and then click "Edit...". Another dialog window will then appear. Press the button marked "New" and enter the path vars for python (C:\Users\<user_name>\AppData\Local\Programs\Python\Python311\). Repeat the same actions for SQLite3 (C:\Users\<user_name>\). Click/save the changes then exit the Settings dialog.

Execute the Install script:

While still in the current Musica directory, cd to the install directory (% cd ./install) then execute musica_db_install.py script. This script will configure the application, setup system paths, and prepare the program to be used:

```
% cd <HOME>
% cd ./Musica/install (or % cd Musica\install on Windows)
% ./musica_db_install.py
```

As part of the install script, the user is prompted for the current (intended) location of the musica.db. The default location is the user's <HOME>²/Musica directly. But the db can also be installed on a shared/network drive by entering the path name (for example, linux would be <PATH>³/Musica. Once the install completes, the Musica/config/musica.conf file is created and populated with path and environment variables specifically tailored to the user's computing environment. Going forward, home (/home/<user_id>) and networked paths (/mnt/<mounted_drive>) will be used interchangeably.

Setup the database schema:

Once the install utility has been executed, it is time to create/setup the database itself. From the Musica directory, execute the following command -

```
% cd <PATH>/Musica/admin
% ./musica_db_sql_provision_sqlite.py
```

The install command will configure the required environment from the data listed in the musica-

2 The <HOME> can be the users home path – I.E. ~/Musica or /home/<user_id>/Musica or C:\Users\<user_id>/Musica (for Windows users)

3 The <PATH> path can be any mounted or share drive – I.E. /mnt/<mounted drive>/Musica or D:\Musica (for Windows users)

conf.template file. The install utility will properly build the musica.conf file that the application uses for data paths, file locations, log files, etc.

Confirm the database:

Once the database is installed, it needs to be validated. From the top-level Musica directory, cd to the admin directory and execute musica_db_debug.py:

```
% cd <PATH>/Musica  
  
    or  
% cd /<network_path>/Musica  
% cd admin  
% ./musica_db_debug.py
```

Once the debug utility has been executed, launch sqlite from the command-line and confirm that both the schema and tables have been loaded:

```
% cd <PATH>/Musica  
  
    or  
% cd /<network_path>/Musica  
% sqlite3 ./musica.db  
% .schema  
% .tables
```

Exit sqlite by entering .quit at the sqlite3 prompt.

Data Rebase:

If you are an existing user, and you saved your current database, you can copy/move the current musica.db to the newly installed Musica directory. A new musica.db will have been created, but it will only contain the database schema and tables. Once the previous musica.db has been copied over to the new install path, the database will need to be rebased in order to work with the new data dictionary.

Navigate to the Musica/admin directory and locate the musica_db_rebase.py utility. The command to invoke the rebase tool requires the current path to the musica.db to be rebased. From the admin directory:

```
% cd <PATH>/Musica  
  
    or  
% cd /<network_path>/Musica  
% cd ./admin  
% ./musica_db_rebase.py ./musica.db
```

The utility will inspect the current db, return the number of records, perform an inspection of the old database and make a pre-base copy before it rebases the musica.db. The rebase utility will return “Rebase OK” upon completion when the rebase executes and exits cleanly. If the rebase fails, it is typically along these lines:

Error category	What it means	Resolution
Path/file	Wrong database supplied or inaccessible	Correct the database path/access
Schema	Not the expected Musica V1 database	Verify database/version; don't force rebase
Database integrity/locking	Database is damaged or in use	Stop and repair/close competing access
Rebase safety failure	Internal validation detected. unexpected change	Do not proceed; use reported backup

If the rebase fails after the backup is created, the utility rolls back the transaction and reports the pre-rebase backup.

At this point, the database is configured and is ready to accept data. (Please refer to the Musica-Notes User Guide on how to load records into the database.)